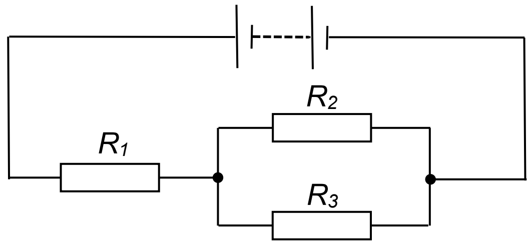
21 The diagram shows a network of three identical resistors connected to a battery of negligible internal resistance.



What is the ratio of $\dfrac{\text{power dissipated in } R_1}{\text{power dissipated in } R_2}$?

A 1 **B** 2 **C** 4 **D** 9

Ans: C

Total resistance of R_2 and $R_3 = (\frac{1}{R}+\frac{1}{R})^{-1} = \left(\frac{2}{R}\right)^{-1} = \frac{1}{2}R$

$$I = \frac{E}{R_{total}} = \frac{E}{R+\frac{1}{2}R}$$

p.d. across R_1, $V_1 = \dfrac{E}{R+\frac{1}{2}R}(R) = \dfrac{2}{3}E$

p.d. across R_2, $V_2 = E - \dfrac{2}{3}E = \dfrac{1}{3}E$

$$\frac{P_1}{P_2} = \frac{\frac{V_1^2}{R}}{\frac{V_2^2}{R}} = \frac{2^2}{1^2} = 4$$

22 A typical potentiometer circuit is shown.